

Neurobiological Foundations of Reading Acceleration: A Naturalistic fNIRS Brain Imaging Study

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METHODS

1. Participants

Thirty-nine children were initially recruited from public primary schools in Rio de Janeiro, Brazil, in grades 3 and 4 (ages 8-9 years). Following neuropsychological screening, 14 children (mean age = 8.85 years, SD = 0.77) met all inclusion criteria: Full-Scale IQ ≥ 85 on the Wechsler Abbreviated Scale of Intelligence (WASI), average or above-average performance on the School Performance Test (TDE) reading and writing subtests, and no reported neurological or psychiatric disorder. Participants were randomly assigned using a computer-generated sequence to one of two groups: the Accelerated group ($n = 7$; 1 female; mean age = 9.00 $\pm$ 0.87 years) and the Non-Accelerated control group ($n = 7$; 3 females; mean age = 8.70 $\pm$ 0.59 years). Written informed consent was obtained from parents or guardians and oral assent from all children.

2. Experimental Protocol and Reading Acceleration Software

ACELETRA(R) is the Brazilian Portuguese adaptation of the Reading Acceleration Programme (RAP), a computer-based intervention employing progressive visual masking to train rapid visual word recognition. On each trial, a sentence appeared on screen and was progressively masked from left to right; participants read the sentence and answered a comprehension question. Presentation speed increased adaptively as performance improved, challenging readers to process text more rapidly while maintaining comprehension accuracy.

The protocol comprised 10 sessions conducted over approximately three weeks (two sessions per week): one calibration session (four blocks x five stimuli) and nine training sessions (seven blocks x five stimuli each; labelled ac_01 to ac_09). Both groups used ACELETRA(R), but only the Accelerated group received the adaptive speed-increase algorithm; the Non-Accelerated group completed matched-difficulty reading without speed escalation. Concurrent fNIRS data were recorded during all sessions in the naturalistic classroom environment.

3. Behavioural Analyses

The primary behavioural outcome was TimeRm (median reading time per character, ms per character; lower values indicate faster reading). A secondary outcome was reading accuracy (number of correct comprehension responses per session). Longitudinal trajectories were modelled using linear mixed models (LMMs) in Python 3.12 (statsmodels 0.14.6, REML):

$$\text{TimeRm} \sim \text{session} * \text{group} + (1 \mid \text{subject})$$

where session (0-9) and group (reference: Non-Accelerated) were fixed effects and a random intercept per subject captured between-subject variability. The same structure was applied to Accuracy.

4. fNIRS Data Acquisition

fNIRS data were recorded using a NIRSport2 8-8 Core Unit (Brain Products GmbH; native sampling rate 4.36 Hz; wavelengths 760 and 850 nm). Twenty-five source-detector channel pairs covered the left hemisphere, targeting two ROIs: Frontal (7 channels: S1-D1, S2-D1, S2-D3, S1-D2, S3-D3, S4-D1, S3-D1) and Temporal (9 channels: S8-D7, S8-D4, S7-D7, S7-D4, S7-D6, S6-D4, S5-D5, S5-D6, S5-D4; channel S6-D5 was excluded due to insufficient signal quality). The portable system permitted recording in the naturalistic classroom environment.

5. Stimulus Event Reconstruction

A custom MATLAB script (contagem_nova.m) reconstructed block-level stimulus events from raw trial-level markers. Rest channels (fewest events) defined block boundaries. Block onset = first reading stimulus within the block; duration = last reading offset minus first onset; amplitude = 1 (boxcar). Automated checks verified block counts (4 for calibration; 7 for training) and stimulus counts (5 per block). A single compound stimulus object ("Aceletra") replaced all original channels.

6. fNIRS Data Preprocessing

Preprocessing used the NIRS Toolbox in MATLAB (pre_processamento.m) through seven sequential steps applied to all 140 sessions (14 participants x 10 sessions): (1) resampling to 4 Hz; (2) baseline trimming (preBaseline = 0 s, postBaseline = 10 s); (3) optical density conversion; (4) modified Beer-Lambert law (yielding HbO, HbR, HbT); (5) total haemoglobin calculation; (6) band-pass filtering (0.01-0.10 Hz); and (7) wavelet-based motion artefact correction. First-level GLM estimation used AddShortSepRegressors = 1 to control for superficial physiological noise, yielding 140 SubjStats objects.

7. Neural Efficiency Index

The neural efficiency (NE) index was computed following Curtin and Ayaz (2019). Mean HbO beta per ROI per participant per session was z-scored separately for each ROI (z_beta). TimeRm was z-scored once over the full training sample (z_TimeRm). The NE

formula was: $NE = (-z_TimeRm - z_beta) / \sqrt{2}$. The negative sign on z_TimeRm ensures that both components increase with better performance (faster reading) and lower activation. Higher NE reflects a more favourable combination of faster reading with lower cortical activation. Calibration-session values provided per-participant baseline covariates ($TimeRm_baseline$, $beta_baseline$). NE was computed for training sessions 1-9 (N = 252: 14 subjects x 9 sessions x 2 ROIs). Separate LMMs per ROI (REML, N = 126 each):

$$NE \sim group * sessao_num + TimeRm_baseline + (1 | subject)$$

8. Cortical Activation (Beta) Analysis

Group-level activation was modelled via a second-level GLM in MATLAB (MixedEffects.m; formula: $beta \sim -1 + group:experiment + (1|subject)$, robust = 1) and a Python LMM on ROI-aggregated HbO betas (all sessions 0-9, N = 280):

$$beta \sim group * sessao_num + roi + (1 | subject) \quad [ML, BFGS]$$

The same structure was applied to StO2 as a supplementary analysis. Mann-Whitney U tests compared mean beta per subject (averaged across training sessions 1-9) between groups for each ROI.

9. Frontal-Temporal Functional Connectivity

Inter-regional connectivity used autoregressive Pearson correlation (ar_corr , model order 4xFs) applied per task block ($divide_events = true$, $min_event_duration = 10$ s, $AddShortSepRegressors = 1$). Mean Fisher-z was averaged over 60 frontal x temporal channel pairs per session. Three sessions were excluded (SUBJ_001, SUBJ_005, SUBJ_024 at session 4) due to signal quality, leaving N = 121. LMM (REML, BFGS):

$$conexao_inter \sim group * session + conn_baseline + (1 | subject)$$

10. Channel-Level Cortical Activation

For each of the 15 HbO channels (S4-D1 excluded), a separate LMM was estimated: $beta \sim group * sessao_num + (1|subject)$ [ML, BFGS]. The group x session interaction coefficient was corrected for multiple comparisons using the Benjamini-Hochberg FDR procedure ($\alpha = 0.05$, 15 tests). Anatomical localisation used the 10-10 system.

11. Correlation Between HbO Slope and Reading Speed Slope

For the six FDR-significant channels, individual OLS slopes were computed (HbO $beta \sim sessao_num$; $TimeRm \sim sessao_num$; sessions 1-9). Spearman correlations between HbO slope and TimeRm slope were computed per channel across three scopes: all subjects (N = 13; SUBJ_030 excluded as outlier), Accelerated (N = 7), and Non-Accelerated (N = 6). FDR correction (Benjamini-Hochberg) was applied within each scope across the six channel tests.

12. General Statistical Considerations

All LMMs included a random intercept per subject. The reference category for group was nao_acelerado (Non-Accelerated) in all models. REML was used for final parameter estimates; ML was used when model structures required likelihood comparisons. The alpha level was set at .05 (two-tailed) for all tests. Influential observations were identified using Cook's distance > 1 and removed when confirmed as data-quality artefacts. Analyses were conducted in Python 3.12 (statsmodels 0.14.6) and MATLAB with the NIRS Toolbox.

RESULTS

1. Reading Speed and Accuracy

TimeRm (ms per character; lower values = faster reading) served as the primary behavioural outcome. Both groups completed all ten sessions. The Accelerated group showed a steeper decline in TimeRm across training sessions relative to the Non-Accelerated group, consistent with the adaptive speed algorithm embedded in ACELETRA(R). Full behavioural LMM results are provided in supplementary materials.

2. Neural Efficiency

For the Frontal ROI ($N = 126$), no fixed effects reached statistical significance. The group $\times$ session interaction was $\beta = 0.067$, $SE = 0.050$, $z = 1.352$, $p = .176$, 95% CI $[-0.030, 0.164]$. TimeRm_baseline was a significant negative predictor ($\beta = -0.012$, $SE = 0.003$, $z = -4.053$, $p < .001$, 95% CI $[-0.018, -0.006]$), indicating that participants with faster baseline reading showed lower NE scores during training, consistent with a ceiling effect in high-performing readers.

For the Temporal ROI ($N = 126$), the group $\times$ session interaction was statistically significant ($\beta = 0.109$, $SE = 0.047$, $z = 2.298$, $p = .022$, 95% CI $[0.016, 0.202]$), indicating that the Accelerated group showed a significantly steeper increase in temporal NE across training sessions relative to the Non-Accelerated group. TimeRm_baseline was also a significant predictor ($\beta = -0.010$, $SE = 0.003$, $z = -2.907$, $p = .004$, 95% CI $[-0.017, -0.003]$).

Mann-Whitney comparisons of mean NE (averaged across training sessions) confirmed significantly higher NE in the Accelerated group in both ROIs: Frontal: $U = 46$, $p = .004$, $r = -0.878$ (large effect); Temporal: $U = 47$, $p = .002$, $r = -0.918$ (large effect).

Table 1. LMM Results for Neural Efficiency

Panel A - Frontal ROI ($N = 126$)

Parameter	Beta	SE	z	p	95% CI
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Intercept	1.131	0.500	2.263	.024	[0.151, 2.110]
Group (Accelerated)	0.091	0.352	0.258	.796	[-0.598, 0.780]
Session	0.046	0.035	1.297	.194	[-0.023, 0.114]
Group x Session	0.067	0.050	1.352	.176	[-0.030, 0.164]
TimeRm_baseline **	-0.012	0.003	-4.053	<.001	[-0.018, -0.006]

Panel B - Temporal ROI (N = 126)

Parameter	Beta	SE	z	p	95% CI
Intercept	0.906	0.562	1.612	.107	[-0.195, 2.007]
Group (Accelerated)	-0.044	0.371	-0.119	.906	[-0.771, 0.683]
Session	0.027	0.034	0.810	.418	[-0.039, 0.093]
Group x Session *	0.109	0.047	2.298	.022	[0.016, 0.202]
TimeRm_baseline **	-0.010	0.003	-2.907	.004	[-0.017, -0.003]

* $p < .05$; ** $p < .001$. REML estimation. Reference group = Non-Accelerated.

3. Cortical Activation (Beta)

The HbO LMM (N = 280) yielded no statistically significant fixed effects. The group x session interaction was $\beta = -0.118$, $SE = 0.115$, $z = -1.034$, $p = .301$, 95% CI [-0.343, 0.106], indicating no differential ROI-level HbO activation trajectory between groups across training. Mann-Whitney tests on mean HbO β confirmed the absence of group differences: Frontal $U = 13$, $p = .165$, $r = 0.469$; Temporal $U = 16$, $p = .318$, $r = 0.347$. The StO₂ supplementary LMM also showed no significant effects (group x session $\beta = -0.017$, $p = .788$).

Table 2. LMM Results - HbO Cortical Activation (N = 280)

Parameter	Beta	SE	z	p	95% CI
Intercept (Non-Acc., Frontal, Session 0)	-0.318	0.517	-0.615	.539	[-1.331, 0.695]
Group (Accelerated)	-0.126	0.693	-0.181	.856	[-1.484, 1.233]
ROI (Temporal vs Frontal)	0.463	0.329	1.408	.159	[-0.182, 1.108]
Session Progression	0.036	0.081	0.443	.658	[-0.123, 0.195]
Group x Session	-0.118	0.115	-1.034	.301	[-0.343, 0.106]

ML; BFGS; Reference: Non-Accelerated, Frontal, Session 0.

4. Frontal-Temporal Functional Connectivity

Mean inter-ROI connectivity (back-transformed to r for descriptive purposes): Accelerated group $M = 0.296$ ($SD = 0.190$), Non-Accelerated group $M = 0.255$ ($SD = 0.149$). The LMM ($N = 121$) revealed no significant group $\times$ session interaction ($\beta = 0.002$, $SE = 0.013$, $z = 0.178$, $p = .859$, 95% CI $[-0.024, 0.028]$). No other fixed effects reached significance. Mann-Whitney comparison of mean per-subject connectivity across training sessions found no group difference ($U = 33$, $p = .318$, $r = -0.347$, medium effect, n.s.).

Table 3. LMM Results - Frontal-Temporal Connectivity (HbO Fisher- z , $N = 121$)

Parameter	Beta	SE	z	p	95% CI
Intercept	0.190	0.086	2.208	.027	[0.021, 0.358]
Group (Accelerated)	0.039	0.112	0.345	.730	[-0.182, 0.259]
Session	0.016	0.009	1.742	.082	[-0.002, 0.034]
Group $\times$ Session	0.002	0.013	0.178	.859	[-0.024, 0.028]
Connectivity Baseline	0.031	0.216	0.142	.887	[-0.393, 0.455]

REML; BFGS; 3 obs. excluded (SUBJ_001, SUBJ_005, SUBJ_024 at session 4).

5. Channel-Level Cortical Activation

Six of the 15 channels tested reached FDR significance ($q < .05$) for the group $\times$ session interaction. In the Frontal ROI, one channel was significant: S3-D1 (L Middle Frontal Gyrus, 66%; 10-10: AF3-F5), $\beta = -0.296$, $p = .003$, $q = .014$, 95% CI $[-0.490, -0.103]$. The negative coefficient indicates that the Accelerated group showed a steeper session-by-session decline in HbO activation at this channel relative to the Non-Accelerated group.

In the Temporal ROI, five channels were FDR-significant. Four showed negative interactions (Accelerated group declining faster): S5-D4 (L Middle Temporal Gyrus, 41%; P7-TP7), $\beta = -0.297$, $p = .002$, $q = .014$, 95% CI $[-0.481, -0.112]$; S7-D7 (L Middle Temporal Gyrus, 55%; T7-C5), $\beta = -0.298$, $p = .002$, $q = .014$, 95% CI $[-0.490, -0.106]$; S5-D6 (L Middle Temporal Gyrus, 47%; P7-P5), $\beta = -0.318$, $p = .006$, $q = .021$, 95% CI $[-0.547, -0.090]$; and S7-D4 (L Middle Temporal Gyrus, 72%; CP5-TP7), $\beta = -0.197$, $p = .009$, $q = .023$, 95% CI $[-0.346, -0.049]$. One channel showed a positive interaction: S5-D5 (L Inferior Temporal Gyrus, 29%; P7-P9), $\beta = +0.215$, $p =$

.007, $q = .021$, 95% CI [0.059, 0.372], indicating a relative increase in HbO in the Accelerated group at this location.

Table 4. Channel-Level LMM - FDR-Significant Channels (HbO)

Channel	10-10	Region	ROI	Beta	p	q(FDR)	95% CI
S3-D1	AF3-F5	L Middle Frontal Gyrus (66%)	Frontal	-0.296	.003	.014	[-0.490, -0.103]
S5-D4	P7-TP7	L Middle Temporal Gyrus (41%)	Temporal	-0.297	.002	.014	[-0.481, -0.112]
S7-D7	T7-C5	L Middle Temporal Gyrus (55%)	Temporal	-0.298	.002	.014	[-0.490, -0.106]
S5-D6	P7-P5	L Middle Temporal Gyrus (47%)	Temporal	-0.318	.006	.021	[-0.547, -0.090]
S5-D5	P7-P9	L Inferior Temporal Gyrus (29%)	Temporal	+0.215	.007	.021	[0.059, 0.372]
S7-D4	CP5-TP7	L Middle Temporal Gyrus (72%)	Temporal	-0.197	.009	.023	[-0.346, -0.049]

FDR: Benjamini-Hochberg; 15 tests; ML estimation; BFGS optimiser.

6. Correlation Between HbO Slope and Reading Speed Slope

Spearman correlations between individual HbO slopes and TimeRm slopes across the six FDR-significant channels did not reach significance after FDR correction in any scope. In the full-sample analysis ($N = 13$), Spearman rho ranged from -0.456 (S7-D4, $p = .117$) to -0.148 (S5-D4, $p = .629$); all $q > .62$. In the Non-Accelerated subgroup ($N = 6$), S7-D4 showed a nominally significant correlation ($\rho = -0.886$, $p = .019$) that did not survive FDR correction ($q = .113$). All Accelerated subgroup correlations were non-significant (all $p > .29$).

Table 5. Individual Neural Efficiency Slopes (NE ~ session, OLS)

Subject	Group	ROI	Slope	p	R2	Sig.
SUBJ 002	Accelerated	Frontal	0.193	0.016	0.590	$p < .05$
SUBJ 002	Accelerated	Temporal	0.106	0.085	0.364	$p < .10$

SUBJ_004	Accelerated	Frontal	0.196	0.229	0.199	n.s.
SUBJ_004	Accelerated	Temporal	0.257	0.001	0.811	p<.05
SUBJ_006	Accelerated	Frontal	-0.012	0.873	0.004	n.s.
SUBJ_006	Accelerated	Temporal	-0.003	0.979	0.000	n.s.
SUBJ_007	Accelerated	Frontal	0.107	0.656	0.030	n.s.
SUBJ_007	Accelerated	Temporal	0.232	0.034	0.497	p<.05
SUBJ_027	Accelerated	Frontal	0.053	0.389	0.108	n.s.
SUBJ_027	Accelerated	Temporal	0.198	0.002	0.755	p<.05
SUBJ_028	Accelerated	Frontal	0.110	0.061	0.415	p<.10
SUBJ_028	Accelerated	Temporal	0.173	0.108	0.327	n.s.
SUBJ_034	Accelerated	Frontal	0.142	0.053	0.437	p<.10
SUBJ_034	Accelerated	Temporal	-0.010	0.949	0.001	n.s.
SUBJ_001	Non-Accelerated	Frontal	0.068	0.354	0.123	n.s.
SUBJ_001	Non-Accelerated	Temporal	0.005	0.920	0.002	n.s.
SUBJ_005	Non-Accelerated	Frontal	0.003	0.945	0.001	n.s.
SUBJ_005	Non-Accelerated	Temporal	0.025	0.738	0.017	n.s.
SUBJ_018	Non-Accelerated	Frontal	0.078	0.138	0.286	n.s.
SUBJ_018	Non-Accelerated	Temporal	0.001	0.993	0.000	n.s.
SUBJ_024	Non-Accelerated	Frontal	0.115	0.204	0.219	n.s.
SUBJ_024	Non-Accelerated	Temporal	0.169	0.001	0.801	p<.05
SUBJ_030	Non-Accelerated	Frontal	-0.116	0.124	0.303	n.s.
SUBJ_030	Non-Accelerated	Temporal	-0.130	0.219	0.207	n.s.
SUBJ_035	Non-Accelerated	Frontal	-0.040	0.515	0.063	n.s.
SUBJ_035	Non-Accelerated	Temporal	-0.049	0.664	0.029	n.s.
SUBJ_037	Non-Accelerated	Frontal	0.210	0.017	0.582	p<.05
SUBJ_037	Non-Accelerated	Temporal	0.169	0.023	0.546	p<.05